Supplementary Material for `Phyloom`: a flexible and extensible framework for simulating phylogenetic trees and networks

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S1 Birth-death-sampling-hybridization models

The birth-death-sampling-hybridization (`bdsh`) model simulates a population of lineages forward in time, from a single ancestor in a given initial state. Every lineage carries a discrete state label, such as a stage of infection, a host type, a location, or a trait. Four event rules act on a single lineage drawn at random, and can be restricted to lineages in a given state: `birth` replaces a lineage by two descendants, each keeping the parent's state or taking a new one; `death` ends a lineage without observing it; `sampling` records a lineage as an observed tip, and either ends it (sampling with removal) or lets it continue (sampling without removal); and `migration` moves a lineage to another state. The three hybridization rules, `ldh`, `lnh`, and `lgh`, act on a pair of lineages drawn at random, at a rate per eligible pair, and join the two parents into a hybrid lineage, differing in whether neither, one, or both parents persist. Scheduled actions change the population at fixed times instead, for instance through `rho_sample`, which samples each current lineage with a fixed probability, as in sampling at the present. The resulting history is a tree, or a network once hybridizations occur.

S1.1 BD: constant rates with a sampling intervention

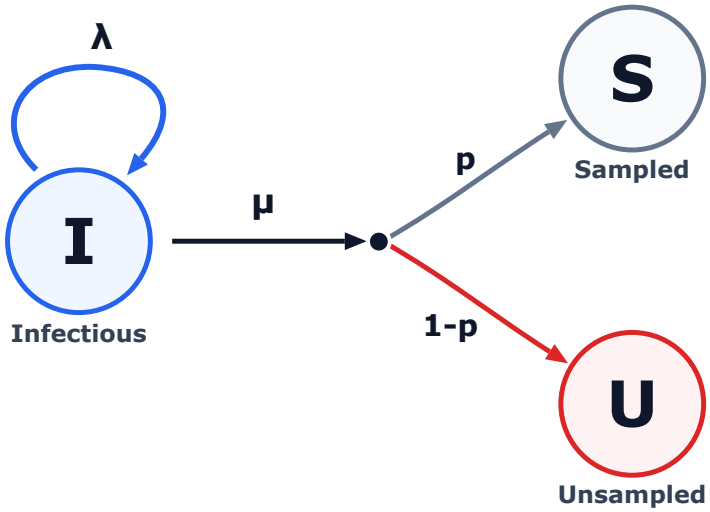


Figure S1: Birth-Death model.

The constant-rate birth-death model with sampling through time [1], the simplest phylodynamic model of an epidemic. Infectious lineages (state `I`) transmit, i.e. branch, at rate λ and are removed (by recovery, death, or isolation) at rate μ; each removal is sampled with probability p. Rather than drawing λ directly, the context draws the basic reproduction number $R_0 = \lambda/\mu$, the expected number of secondary infections caused by one infected individual [2], and the mean infectious period $1/\mu$, and derives $\lambda = R_0\mu$. Removal at total rate μ is written as two competing events, unsampled death at rate $\mu(1-p)$ and sampling at rate μp. Sampling effort rises at a random intervention time, as when testing is scaled up during an outbreak: p goes from `p_before` to `p_after`, at least twice as large, and both removal rates are `skyline` hazards [3] that switch at `intervention_time` and always sum to μ, so the intervention changes which removals are observed, not the epidemic dynamics. Each trial stops when `n_leaves` tips have been sampled, and is kept only if the epidemic is still ongoing and at least one tip was sampled before the intervention, so that the tree informs `p_before`. Each sample stores the full history (`history.nwk`) and the sampled tree (`sampled.nwk`); `metadata.csv` also records the number of tips sampled before the intervention.

```
output_dir: outputs/BD-yaml
n_samples: {train: 100, val: 10, test: 10}
n_jobs: -1
seed: 42
```

```
context:
  R_0: {type: lognormal, mean: 1.0, sigma: 0.2}
  mu: {type: uniform, low: 0.25, high: 0.6}
  p_before: {type: uniform, low: 0.05, high: 0.15}
  p_after: {type: uniform, low: 2 * p_before, high: 0.5}
  la: R_0 * mu
  psi: mu * np.array([p_before, p_after])
  death_rate: mu - psi
  intervention_time: {type: uniform, low: 4, high: 6}
  n_leaves: {type: integers, low: 20, high: 41}

model:
  spec: bdsh
  init_state: I
  events:
    - rule: {spec: birth}
      hazard: {spec: constant, rate: la}
    - rule: {spec: death}
      hazard: {spec: skyline, rates: death_rate, break_times: [intervention_time]}
    - rule: {spec: sampling, removal: true}
      hazard: {spec: skyline, rates: psi, break_times: [intervention_time]}

stop_criterion: model.count_sampled_lineages() == n_leaves

acceptance_criterion: (
    model.count_current_lineages() > 0
    and any(
      l
      for l in model.get_sampled_lineages()
      if model.time_of(l) < intervention_time
    )
  )

logs:
  n_sampled_before_intervention: sum(
      1
      for l in model.get_sampled_lineages()
      if model.time_of(l) < intervention_time
    )

output_hooks:
  - model.history().dump(output_dir / "history.nwk")
  - model.history(model.get_sampled_lineages()).dump(output_dir / "sampled.nwk")
```

S1.2 BDEI: an exposed, non-infectious stage

The birth-death exposed-infectious (BDEI) model [4] adds an incubation period to BD, as for pathogens such as Ebola or SARS-CoV-2, where infected individuals are not infectious straight away. Newly infected lineages start exposed (state `E`): they neither transmit nor are removed, and become infectious (state `I`) at rate ε. Infectious lineages transmit at rate λ, each transmission creating a new `E` lineage beside the parent, and are removed at rate μ, each removal sampled with probability p. The `child_states` of the `birth` rule and the `migration` rule from `E` to `I` are the whole of the difference from BD. This makes BDEI a multi-type birth-death model, in which every rate may depend on the state of the lineage [5, 6].

The context draws the rates through interpretable quantities: R_0, the total duration of infection $T = 1/\varepsilon + 1/\mu$, and the fraction f of it spent incubating, so that the incubation period is $T_E = fT$, the infectious period is $T_I = (1 - f)T$, and $\lambda = R_0/T_I$. The epidemic starts from an exposed individual with probability 0.8. Each trial stops when `n_leaves` tips have been sampled, and is kept only if the epidemic is still ongoing. Each sample stores the sampled tree (`sampled.nwk`);

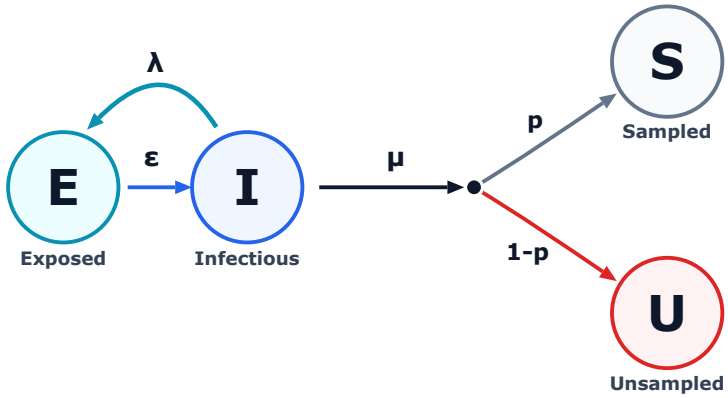


Figure S2: Birth-Death Exposed-Infectious model.

`metadata.csv` also records how many lineages are still incubating or infectious when the trial stops, i.e. the unobserved epidemic behind the tree.

```
output_dir: outputs/BDEI-yaml
n_samples: {train: 100, val: 10, test: 10}
n_jobs: -1
seed: 42

context:
  R_0: {type: lognormal, mean: 1.0, sigma: 0.2}
  T: {type: lognormal, mean: 1.0, sigma: 0.3}
  f: {type: uniform, low: 0.05, high: 0.75}
  T_I: (1 - f) * T
  T_E: f * T
  epsilon: 1 / T_E
  mu: 1 / T_I
  p: {type: uniform, low: 0.05, high: 0.75}
  la: R_0 * mu
  psi: mu * p
  death_rate: mu - psi
  init_state: {type: choice, a: "['E', 'I']", p: "[0.8, 0.2]"}
  n_leaves: {type: integers, low: 50, high: 251}

model:
  spec: bdsh
  init_state: "{init_state}"
  events:
    - rule: {spec: birth, parent_state: I, child_states: [I, E]}
      hazard: {spec: constant, rate: la}
    - rule: {spec: migration, source_state: E, target_state: I}
      hazard: {spec: constant, rate: epsilon}
    - rule: {spec: death, state: I}
      hazard: {spec: constant, rate: death_rate}
    - rule: {spec: sampling, state: I, removal: true}
      hazard: {spec: constant, rate: psi}

stop_criterion: model.count_sampled_lineages() == n_leaves

acceptance_criterion: model.count_current_lineages() > 0

logs:
  still_incubating: model.count_current_lineages("E")
  still_infectious: model.count_current_lineages("I")

output_hooks:
  - model.history(model.get_sampled_lineages()).dump(output_dir / "sampled.nwk")
```

S1.3 BDSS: superspreading lineages

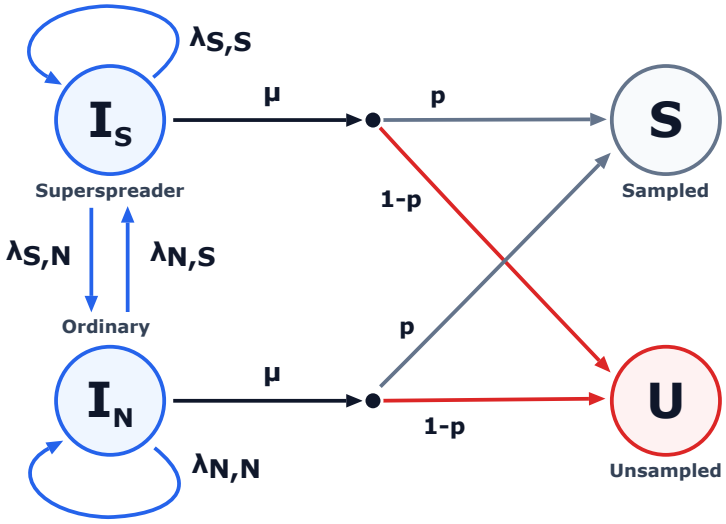


Figure S3: Birth-Death super-spreading model.

The birth-death superspreading (BDSS) model [5] captures heterogeneity in transmission, where a minority of individuals cause a majority of infections. Infectious lineages are either ordinary (`I_N`) or superspreaders (`I_S`), and each class can transmit to either class, giving four birth rates λ_{NN}, λ_{NS}, λ_{SN}, and λ_{SS} (parent class first, offspring class second). Both classes are removed at the same rate $\mu = 1/T_I$, with $T_I = 2.5$, and sampled with probability $p = 0.25$.

The context derives those four rates from three interpretable quantities: the average reproduction number R_0; the superspreading fraction f_{SS}, the probability that a new infection is a superspreader, whatever the class of its infector; and the transmission ratio r_{SS}, the factor by which superspreaders transmit more than ordinary spreaders. Writing $R_{XY} = \lambda_{XY}/\mu$, the expressions in the context solve $R_{NS}/(R_{NN} + R_{NS}) = R_{SS}/(R_{SN} + R_{SS}) = f_{SS}$, $R_{SN} + R_{SS} = r_{SS}(R_{NN} + R_{NS})$, and $(1 - f_{SS})(R_{NN} + R_{NS}) + f_{SS}(R_{SN} + R_{SS}) = R_0$. This kind of reparameterization keeps priors on quantities one can reason about. The first lineage is a superspreader with probability f_{SS}. Each trial stops when `n_leaves` tips have been sampled, and is kept only if the epidemic is still ongoing. Each sample stores the sampled tree (`sampled.nwk`); `metadata.csv` also records the frequency of superspreaders among the lineages still infectious at the end of the trial, which approaches f_{SS} in large outbreaks.

```
output_dir: outputs/BDSS-yaml
n_samples: {train: 100, val: 10, test: 10}
n_jobs: -1
seed: 42

context:
  R_0: {type: lognormal, mean: 1.0, sigma: 0.2}
  T_I: 2.5
  mu: 1 / T_I
  p: 0.25
  psi: mu * p
  death_rate: mu - psi
  r_SS: {type: uniform, low: 2, high: 10}
  f_SS: {type: uniform, low: 0.1, high: 0.3}
  R_0_NS: R_0 * f_SS / (1 + r_SS * f_SS - f_SS)
  R_0_SN: (R_0 - r_SS * R_0_NS) * r_SS
  R_0_S: r_SS * R_0_NS
  R_0_N: R_0_SN / r_SS
  lambda_NN: R_0_N * mu
  lambda_SS: R_0_S * mu
```

```
  lambda_NS: R_0_NS * mu
  lambda_SN: R_0_SN * mu
  init_state: {type: choice, a: "['I_N', 'I_S']", p: "[1 - f_SS, f_SS]"}
  n_leaves: {type: integers, low: 50, high: 251}

model:
  spec: bdsh
  init_state: "{init_state}"
  events:
    - rule: {spec: birth, parent_state: I_N}
      hazard: {spec: constant, rate: lambda_NN}
    - rule: {spec: birth, parent_state: I_S}
      hazard: {spec: constant, rate: lambda_SS}
    - rule: {spec: birth, parent_state: I_N, child_states: [I_N, I_S]}
      hazard: {spec: constant, rate: lambda_NS}
    - rule: {spec: birth, parent_state: I_S, child_states: [I_S, I_N]}
      hazard: {spec: constant, rate: lambda_SN}
    - rule: {spec: death}
      hazard: {spec: constant, rate: death_rate}
    - rule: {spec: sampling, removal: true}
      hazard: {spec: constant, rate: psi}

stop_criterion: model.count_sampled_lineages() == n_leaves

acceptance_criterion: model.count_current_lineages() > 0

logs:
  ss_frequency: model.count_current_lineages("I_S") / model.count_current_lineages()

output_hooks:
  - model.history(model.get_sampled_lineages()).dump(output_dir / "sampled.nwk")
```

S1.4 BiSSE: a binary trait driving speciation and extinction

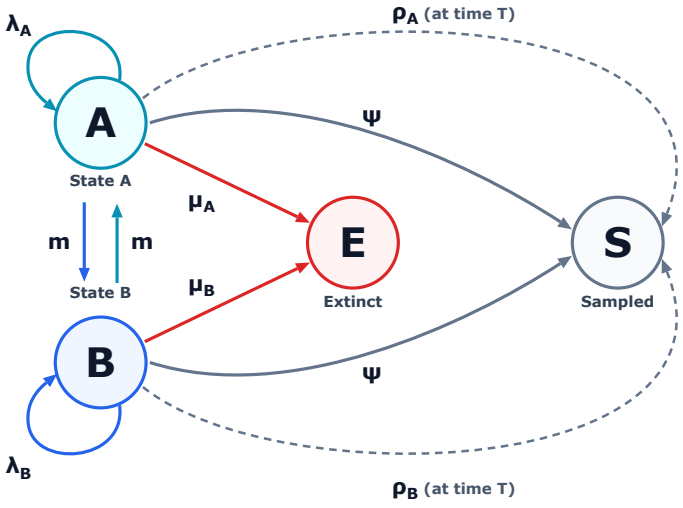


Figure S4: BiSSE model.

The binary-state speciation and extinction (BiSSE) model [7] asks whether a trait, such as a morphological character or a mode of reproduction, affects the rates at which species diversify. A binary trait, `A` or `B`, sets each lineage's speciation (λ_A, λ_B) and extinction (μ_A, μ_B) rates, and lineages switch trait at rate m in either direction. The priors make `B` speciate up to twice as fast as `A`, and keep extinction between a fifth and a half of speciation in both states. The clade is observed in two ways, as in the fossilized birth-death process [8, 9]: fossil-like sampling at rate ψ during

the simulation, with `removal: false` so the lineage survives being sampled and can appear as a sampled ancestor, and a scheduled action at the present (`max_time`) that samples a fraction ρ_A or ρ_B of the surviving lineages of each state. There is no `stop_criterion`, so every trial runs to `max_time` unless the clade dies out first, and trees vary in size. A trial is kept only if more than ten tips of each state were sampled. If the parameters of a sample make this too unlikely, `max_rejections` replaces them with a new draw after ten rejected trials, instead of retrying forever. Each sample stores the sampled tree (`sampled.nwk`), with both fossil-like and present-day samples, whose node labels carry the trait through the tree.

```yaml
output_dir: outputs/BiSSE-yaml
n_samples: {train: 100, val: 10, test: 10}
n_jobs: -1
seed: 42

context:
  la_A: {type: uniform, low: 0.1, high: 0.3}
  la_B: {type: uniform, low: la_A, high: 2 * la_A}
  mu_A: {type: uniform, low: la_A / 5, high: la_A / 2}
  mu_B: {type: uniform, low: la_B / 5, high: la_B / 2}
  psi: {type: uniform, low: 0.001, high: 0.005}
  m: {type: uniform, low: 0.01, high: 0.05}
  init_state: {type: choice, a: "['A', 'B']", p: "[0.5, 0.5]"}
  max_time: {type: uniform, low: 10, high: 20}
  rho_sample_A: {type: uniform, low: 0.5, high: 0.9}
  rho_sample_B: {type: uniform, low: 0.5, high: 0.9}

model:
  spec: bdsh
  init_state: "{init_state}"
  events:
    - rule: {spec: birth, parent_state: A}
      hazard: {spec: constant, rate: la_A}
    - rule: {spec: birth, parent_state: B}
      hazard: {spec: constant, rate: la_B}
    - rule: {spec: death, state: A}
      hazard: {spec: constant, rate: mu_A}
    - rule: {spec: death, state: B}
      hazard: {spec: constant, rate: mu_B}
    - rule: {spec: migration, source_state: A, target_state: B}
      hazard: {spec: constant, rate: m}
    - rule: {spec: migration, source_state: B, target_state: A}
      hazard: {spec: constant, rate: m}
    - rule: {spec: sampling, removal: false}
      hazard: {spec: constant, rate: psi}
  schedule:
    - times: [max_time]
      action: |
        model.rho_sample(rho_sample_A, "A", removal=False)
        model.rho_sample(rho_sample_B, "B", removal=False)

max_time: max_time

acceptance_criterion: (
    model.count_sampled_lineages("A") > 10
    and model.count_sampled_lineages("B") > 10
  )

max_rejections: 10

output_hooks:
  - model.history(model.get_sampled_lineages()).dump(output_dir / "sampled.nwk")
```

S1.5 BDSH: three kinds of hybridization

This example generates phylogenetic networks under a birth-death process with hybridization [10, 11]. Hybridization rules act on a pair of lineages and join them into a hybrid lineage with two parents, so the history is a network rather than a tree. Lineages branch at rate λ and are sampled at rate ψ without being removed, while pairs of lineages hybridize through three competing events [12]: lineage-degenerative hybridization (`ldh`), in which the two parents merge into a single hybrid child, so the number of lineages decreases by one; lineage-neutral hybridization (`lnh`), in which one parent persists while the other is absorbed into the hybrid child, so the number of lineages is unchanged; and lineage-generative hybridization (`lgh`), in which both parents persist and the hybrid is a new lineage, so the number of lineages increases by one. Each kind of hybrid gets its own state, so the output records which kind of event produced each reticulation. Hybridization rules act on lineage *pairs* rather than lineages, so their total rate grows quadratically with the size of the clade. Each rate is drawn independently for every sample, so the dataset covers networks dominated by one kind of hybridization as well as mixtures.

Each trial stops when `n_leaves` tips have been sampled. A trial is accepted only if the sampled network really contains a reticulation, that is, if it has more edges than a tree on the same nodes would (a tree has one edge fewer than nodes). A hybridization disappears from the sampled network when one of its parents has no other sampled descendants. Each sample stores the full history (`history.nwk`) and the sampled network (`sampled.nwk`), in extended Newick; `metadata.csv` also records the number of tips and reticulations visible in the sampled network.

```
output_dir: outputs/BDSH-yaml
n_samples: {train: 10, val: 2, test: 2}
n_jobs: -1
seed: 45

context:
  la: 0.5
  sampling_rate: 0.25
  n_leaves: 50
  ldh_rate: {type: uniform, low: 0.0005, high: 0.003}
  lnh_rate: {type: uniform, low: 0.0005, high: 0.003}
  lgh_rate: {type: uniform, low: 0.0005, high: 0.003}

model:
  spec: bdsh
  init_state: A
  events:
    - rule: {spec: birth}
      hazard: {spec: constant, rate: la}
    - rule: {spec: sampling, removal: false}
      hazard: {spec: constant, rate: sampling_rate}
    - rule: {spec: ldh, hybrid_state: H_LDH}
      hazard: {spec: constant, rate: ldh_rate}
    - rule: {spec: lnh, hybrid_state: H_LNH}
      hazard: {spec: constant, rate: lnh_rate}
    - rule: {spec: lgh, hybrid_state: H_LGH}
      hazard: {spec: constant, rate: lgh_rate}

stop_criterion: model.count_sampled_lineages() == n_leaves

acceptance_criterion: (
    model.history(model.get_sampled_lineages()).n_edges
    > model.history(model.get_sampled_lineages()).n_nodes - 1
  )

logs:
  sampled_tips: model.history(model.get_sampled_lineages()).count_leaves()
  reticulations: (
      model.history(model.get_sampled_lineages()).n_edges
```

```
      - model.history(model.get_sampled_lineages()).n_nodes
      + 1
    )

output_hooks:
  - model.history().dump(output_dir / "history.nwk")
  - model.history(model.get_sampled_lineages()).dump(output_dir / "sampled.nwk")
```

S1.6 SeasonalBD: a seasonal transmission rate

Many infections, such as influenza, are transmitted more efficiently in some seasons than in others. This is a birth-death process whose transmission rate is a function of time [13], here a seasonal cycle,

$$\lambda(t) = \lambda_0 \left(1 + a \cos\left(\frac{2\pi(t - t_p)}{P}\right)\right),$$

with mean $\lambda_0 = R_0\mu$, amplitude a, period P (one time unit, e.g. a year), and peak at t_p. Lineages are removed at a constant rate μ and each removal is sampled with probability p, so only transmission is seasonal.

The built-in hazards are constant, piecewise constant (`skyline`), exponential, or a point mass (`delta`), and none of them covers this rate. It is therefore given as a plain expression of the simulated time `t` through the `numeric` hazard, which integrates it by quadrature and inverts the cumulative hazard numerically; firing times are exact up to the tolerance of the solver, with no discretization of time. `initial_dt` is where the root solver starts bracketing a firing time, and setting it to a typical waiting time, here $1/\lambda_0$, keeps that search short. Each trial stops when `n_leaves` tips have been sampled, and is kept only if the epidemic is still ongoing. Each sample stores the full history (`history.nwk`) and the sampled tree (`sampled.nwk`); `metadata.csv` also records how many tips were sampled while transmission was above its mean, that is, within a quarter period of a seasonal peak.

```
output_dir: outputs/SeasonalBD-yaml
n_samples: {train: 100, val: 10, test: 10}
n_jobs: -1
seed: 42

context:
  R_0: {type: lognormal, mean: 1.0, sigma: 0.2}
  mu: {type: uniform, low: 0.25, high: 0.6}
  la_0: R_0 * mu
  amplitude: {type: uniform, low: 0.3, high: 0.9}
  period: 1.0
  peak_time: {type: uniform, low: 0.0, high: 1.0}
  p: {type: uniform, low: 0.1, high: 0.4}
  psi: mu * p
  death_rate: mu - psi
  n_leaves: {type: integers, low: 20, high: 41}

model:
  spec: bdsh
  init_state: I
  events:
    - rule: {spec: birth}
      hazard:
        spec: numeric
        rate_fn: la_0 * (1 + amplitude * np.cos(2 * np.pi * (t - peak_time) / period))
        initial_dt: 1 / la_0
    - rule: {spec: death}
      hazard: {spec: constant, rate: death_rate}
    - rule: {spec: sampling, removal: true}
      hazard: {spec: constant, rate: psi}
```

```
stop_criterion: model.count_sampled_lineages() == n_leaves

acceptance_criterion: model.count_current_lineages() > 0

logs:
  n_sampled_in_high_season: sum(
      1
      for l in model.get_sampled_lineages()
      if np.cos(2 * np.pi * (model.time_of(l) - peak_time) / period) > 0
    )

output_hooks:
  - model.history().dump(output_dir / "history.nwk")
  - model.history(model.get_sampled_lineages()).dump(output_dir / "sampled.nwk")
```

S1.7 DiversityDependent: diversity dependence and competitive replacement

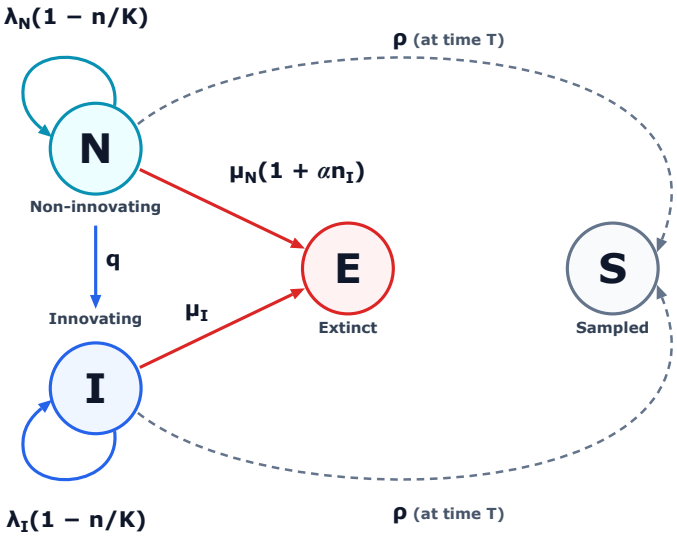


Figure S5: Diversity-dependent model.

In diversity-dependent models [14, 15], speciation slows as a clade fills its ecological niches, so diversity saturates. This example combines diversity dependence with competitive replacement. A clade of non-innovating lineages (state `N`) diversifies until it approaches a carrying capacity K. `N` lineages acquire a key innovation at rate q and become innovating lineages (state `I`), which speciate r_I times faster. With n extant lineages, of which n_I are innovators, the speciation rate of each state is multiplied by $\max(0, 1 - n/K)$, so speciation stops at K lineages, and the extinction rate of `N` lineages by $1 + \alpha n_I$, so innovators progressively replace incumbents, with α setting the strength of competition. Hazards depend only on time, so both multipliers are `rate_multiplier` expressions of the live model, evaluated again before every step so that the rates follow the state as it changes. There is no `stop_criterion`, so every trial runs to `max_time` unless the clade dies out first. At the present, a scheduled action samples a fraction ρ of the surviving lineages, and a trial is kept only if at least 20 tips, 5 of them innovators, were sampled. Each sample stores the full history (`history.nwk`), which also shows the extinct incumbents, and the sampled tree (`sampled.nwk`); `metadata.csv` also records the number of extant lineages of each state at the present, which shows how far the replacement has progressed.

```
output_dir: outputs/DiversityDependent-yaml
n_samples: {train: 100, val: 10, test: 10}
n_jobs: -1
```

```yaml
seed: 42

context:
  K: {type: integers, low: 50, high: 101}
  la_N: {type: uniform, low: 0.2, high: 0.4}
  mu_N: {type: uniform, low: 0.02, high: 0.05}
  r_I: {type: uniform, low: 1.5, high: 3}
  la_I: r_I * la_N
  mu_I: {type: uniform, low: 0.02, high: 0.05}
  q: {type: uniform, low: 0.002, high: 0.01}
  alpha: {type: uniform, low: 0.02, high: 0.1}
  max_time: {type: uniform, low: 40, high: 60}
  rho: {type: uniform, low: 0.5, high: 1.0}

model:
  spec: bdsh
  init_state: N
  events:
    - rule: {spec: birth, parent_state: N}
      hazard: {spec: constant, rate: la_N}
      rate_multiplier: max(0.0, 1 - model.count_current_lineages() / K)
    - rule: {spec: birth, parent_state: I}
      hazard: {spec: constant, rate: la_I}
      rate_multiplier: max(0.0, 1 - model.count_current_lineages() / K)
    - rule: {spec: death, state: N}
      hazard: {spec: constant, rate: mu_N}
      rate_multiplier: 1 + alpha * model.count_current_lineages("I")
    - rule: {spec: death, state: I}
      hazard: {spec: constant, rate: mu_I}
    - rule: {spec: migration, source_state: N, target_state: I}
      hazard: {spec: constant, rate: q}
  schedule:
    - times: [max_time]
      action: |
        model.rho_sample(rho, "N", removal=False)
        model.rho_sample(rho, "I", removal=False)

max_time: max_time

acceptance_criterion: (
    model.count_sampled_lineages() >= 20
    and model.count_sampled_lineages("I") >= 5
  )

logs:
  extant_N: model.count_current_lineages("N")
  extant_I: model.count_current_lineages("I")

output_hooks:
  - model.history().dump(output_dir / "history.nwk")
  - model.history(model.get_sampled_lineages()).dump(output_dir / "sampled.nwk")
```

S1.8 BDAliSim: alignments along birth-death trees

This example requires IQ-TREE 3 [16], which provides AliSim [17], installed with its `iqtree3` command on your `PATH`. It simulates sampled trees under the constant-rate birth-death model with sampling through time [1], with the context drawing R_0, the removal rate μ, and the sampling probability p. Each trial stops when `n_leaves` tips have been sampled, and is kept only if the epidemic is still ongoing. A post command, run last for each sample, then passes the sampled tree to AliSim, which simulates a DNA alignment along it under the Jukes-Cantor model [18]. Branch lengths, in units of time, are converted into substitutions per site by `--branch-scale`, with a strict

clock rate drawn for every sample, and the length of the alignment is drawn uniformly between 500 and 1000 sites. Each sample stores the sampled tree (`sampled.nwk`), the simulated alignment in FASTA format (`alignment.fa`), and the AliSim log (`sampled.nwk.log`).

```
output_dir: outputs/BDAliSim-yaml
n_samples: {train: 100, val: 10, test: 10}
n_jobs: -1
seed: 42

context:
  R_0: {type: uniform, low: 1.5, high: 4.0}
  mu: {type: uniform, low: 0.25, high: 0.6}
  p: {type: uniform, low: 0.1, high: 0.5}
  la: R_0 * mu
  death_rate: mu * (1 - p)
  psi: mu * p
  n_leaves: {type: integers, low: 20, high: 41}
  clock_rate: {type: uniform, low: 0.001, high: 0.01}
  seq_length: {type: integers, low: 500, high: 1001}
  alisim_seed: {type: integers, low: 1, high: 2**31}

model:
  spec: bdsh
  init_state: I
  events:
    - rule: {spec: birth}
      hazard: {spec: constant, rate: la}
    - rule: {spec: death}
      hazard: {spec: constant, rate: death_rate}
    - rule: {spec: sampling, removal: true}
      hazard: {spec: constant, rate: psi}

stop_criterion: model.count_sampled_lineages() == n_leaves

acceptance_criterion: model.count_current_lineages() > 0

output_hooks:
  - model.history(model.get_sampled_lineages()).dump(output_dir / "sampled.nwk")

post_commands:
  - iqtree3 --alisim {output_dir}/alignment -t {output_dir}/sampled.nwk
    -m JC --length {seq_length} --branch-scale {clock_rate}
    --seed {alisim_seed} -af fasta
```

S2 The coalescent

Coalescent models describe the genealogy of a sample backward in time, from the sampled lineages to their most recent common ancestor [19], without simulating the rest of the population. The `coalescent` model starts from an initial set of sampled lineages per state, such as demes, and has two event rules, both of which can be restricted to lineages in given states: `coalesce` merges a pair of lineages into their common ancestor, at a rate per eligible pair that is inversely proportional to the effective population size, so that changes in population size are expressed through a time-varying rate; and `retrace` moves a single lineage to another state, as a migration traced backward in time. Samples taken further in the past are added by scheduled actions that call `spawn`, which adds a new sampled lineage at the scheduled time, giving serially sampled tips.

S2.1 Coalescent: a single population of constant size

The Kingman coalescent [19] describes the genealogy of a sample of lineages drawn from a large, well-mixed population of constant effective size N_e, and is simulated backward in time, from the

samples to their most recent common ancestor. Between 8 and 12 lineages are sampled at the present. Each pair of lineages merges at rate $1/(2N_e)$, with time measured in generations, and counting every pair supplies the $\binom{n}{2}$ factor of the Kingman coalescent, so the total coalescence rate falls from $\binom{n}{2}/(2N_e)$ as lineages merge. The simulation stops when a single lineage, the root, is left. Each sample stores the genealogy (`genealogy.nwk`).

```yaml
output_dir: outputs/Coalescent-yaml
n_samples: {train: 10, val: 2, test: 2}
n_jobs: -1
seed: 42

context:
  n_tips: {type: integers, low: 8, high: 13}
  effective_population_size: 100
  coalescence_rate: 1 / (2 * effective_population_size)

model:
  spec: coalescent
  init_population: {A: n_tips}
  events:
    - rule: {spec: coalesce, current_states: [A, A], ancestral_state: A}
      hazard: {spec: constant, rate: coalescence_rate}

stop_criterion: model.count_current_lineages() == 1

output_hooks:
  - model.history().dump(output_dir / "genealogy.nwk")
```

S2.2 SerialCoalescent: serially sampled tips

In many datasets, samples are not all collected at the same time: ancient DNA, or pathogens sequenced throughout an epidemic, give serially sampled, or heterochronous, tips [20]. Backward in time, such a sample enters the genealogy at its sampling time, and from then on can only coalesce with lineages that are still being traced. Here, between 4 and 8 lineages are sampled at the present, and between 4 and 8 further lineages at times drawn uniformly between the present and $2N_e$ generations in the past, within the expected depth of the genealogy. Each of them is added at its sampling time by a scheduled action that calls `spawn`. The genealogy may coalesce into a single lineage before all samples are added, so the stop criterion also requires the oldest sample to have been added; until then, the single remaining lineage is simply traced further back. Each sample stores the genealogy (`genealogy.nwk`), whose tips lie at their sampling times; `metadata.csv` also records the time to the most recent common ancestor.

```yaml
output_dir: outputs/SerialCoalescent-yaml
n_samples: {train: 10, val: 2, test: 2}
n_jobs: -1
seed: 46

context:
  n_present: {type: integers, low: 4, high: 9}
  n_past: {type: integers, low: 4, high: 9}
  effective_population_size: 100
  coalescence_rate: 1 / (2 * effective_population_size)
  sampling_times: {type: uniform, low: 0, high: 2 * effective_population_size, size: n_past}
  oldest_sampling_time: sampling_times.max()

model:
  spec: coalescent
  init_population: {A: n_present}
  events:
    - rule: {spec: coalesce, current_states: [A, A], ancestral_state: A}
      hazard: {spec: constant, rate: coalescence_rate}
```

```
  schedule:
    - times: sampling_times
      action: model.spawn("A")

stop_criterion: (
    model.current_time >= oldest_sampling_time
    and model.count_current_lineages() == 1
  )

logs:
  tmrca: model.current_time

output_hooks:
  - model.history().dump(output_dir / "genealogy.nwk")
```

S2.3 StructuredCoalescent: two demes connected by migration

The structured coalescent [21, 22] extends the coalescent to a population subdivided into demes, such as geographic locations or host species, between which individuals migrate. Four lineages are sampled from each of two demes, `A` and `B`, of the same effective size, connected by symmetric migration. Backward in time, lineages coalesce only within their deme, at the same pairwise rate as in a single population, and `retrace` rules move them between demes at a constant rate, so two lineages in different demes must come to share a deme before they can merge. As time runs backward, a `retrace` from `A` to `B` means that the ancestor belonged to deme `B`: forward in time, a migration from `B` into `A`. The stop criterion is essential here, since `retrace` events would otherwise keep firing on the root forever. Each sample stores the genealogy (`genealogy.nwk`), in which each `retrace` is kept as a node with one child, and node labels carry the deme, recording the deme of each ancestral lineage over time.

```
output_dir: outputs/StructuredCoalescent-yaml
n_samples: {train: 10, val: 2, test: 2}
n_jobs: -1
seed: 43

context:
  effective_population_size: 1
  coalescence_rate: 1 / (2 * effective_population_size)
  migration_rate: 0.5

model:
  spec: coalescent
  init_population: {A: 4, B: 4}
  events:
    - rule: {spec: retrace, current_state: A, ancestral_state: B}
      hazard: {spec: constant, rate: migration_rate}
    - rule: {spec: retrace, current_state: B, ancestral_state: A}
      hazard: {spec: constant, rate: migration_rate}
    - rule: {spec: coalesce, current_states: [A, A], ancestral_state: A}
      hazard: {spec: constant, rate: coalescence_rate}
    - rule: {spec: coalesce, current_states: [B, B], ancestral_state: B}
      hazard: {spec: constant, rate: coalescence_rate}

stop_criterion: model.count_current_lineages() == 1

output_hooks:
  - model.history().dump(output_dir / "genealogy.nwk")
```

S2.4 ExponentialGrowthCoalescent: an exponentially growing population

A coalescent with a time-varying effective population size [23, 24], here for a population, such as an expanding epidemic, that has grown exponentially at rate g up to the present size $N_e(0)$. Backward in time, the effective size $N_e(t) = N_e(0)\, e^{-gt}$ shrinks and the pairwise coalescence rate $1/(2N_e(t))$ grows as e^{gt}. That is an `exponential` hazard with a negative `decay`. Because coalescence is rare near the present and fast in the small ancestral population, the genealogies are star-like, with long terminal branches and ancestors compressed near the root. The simulation stops when a single lineage, the root, is left. Each sample stores the genealogy (`genealogy.nwk`).

```
output_dir: outputs/ExponentialGrowthCoalescent-yaml
n_samples: {train: 10, val: 2, test: 2}
n_jobs: -1
seed: 44

context:
  n_tips: {type: integers, low: 8, high: 13}
  present_effective_population_size: 1000
  growth_rate: 0.1
  coalescence_rate_at_present: 1 / (2 * present_effective_population_size)
  hazard_decay: -growth_rate

model:
  spec: coalescent
  init_population: {A: n_tips}
  events:
    - rule: {spec: coalesce}
      hazard:
        spec: exponential
        rate: coalescence_rate_at_present
        decay: hazard_decay

stop_criterion: model.count_current_lineages() == 1

output_hooks:
  - model.history().dump(output_dir / "genealogy.nwk")
```

S3 Compartmental epidemics

Compartmental models describe an epidemic in a finite host population [25, 26], so that transmission slows down as susceptible hosts are depleted. The `epidemic` model keeps uninfected compartments, such as susceptible and recovered hosts, as pools of plain counts, and infected hosts as lineages, so that only the transmission tree is recorded in full. Its event rules act on lineages, on hosts in a pool, or on both: `infection` takes a host from a pool, e.g. susceptible, and adds a new lineage beside its infector, at a rate per pair of an infected lineage and a host in that pool, so that transmission slows as the pool empties; `recovery` ends a lineage, optionally sampling it, and returns the host to a pool, e.g. recovered; `migration` moves a lineage to another state, e.g. from exposed to infectious; and `flow` moves a host from one pool to another, e.g. when immunity wanes, at a rate per host in the source pool. The rules that act on lineages can be restricted to lineages in a given state.

S3.1 SIR: susceptible, infectious, recovered

The susceptible-infectious-recovered (SIR) model [25, 26] describes an epidemic in a closed host population, which slows down and eventually burns out as susceptible hosts are depleted. Here it produces a transmission tree, as in phylodynamic analyses of density-dependent transmission [27, 28]. Susceptible (`S`) and recovered (`R`) hosts are pools, counted but not tracked individually; infectious hosts are lineages in state `I`. An infection consumes one susceptible host and creates a new `I` lineage, and an `I` lineage recovers into `R`, either unobserved at rate μ or sampled at rate ψ, so that a fraction

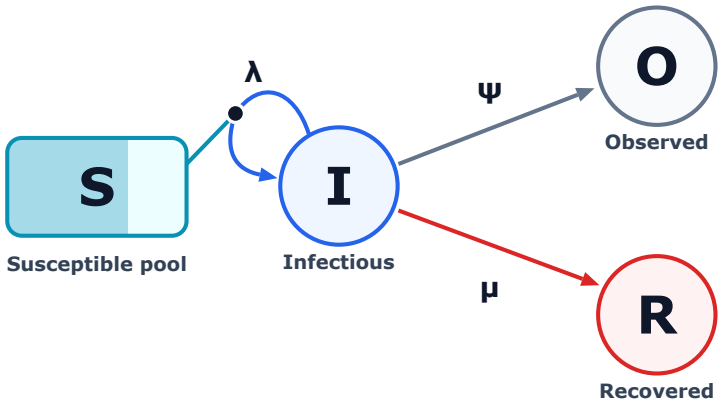


Figure S6: SIR model.

s of recoveries are sampled. The `infection` rule counts every pair of an infectious lineage and a susceptible host, so the per-lineage transmission rate is λS and decreases as the epidemic depletes S. The context sets $\lambda = R_0(\mu + \psi)/S_0$, so that R_0 is the reproduction number at the start of the epidemic, in a population of S_0 susceptible hosts. The trial stops at the twentieth sample, so the trees mostly describe the growth phase, and is kept only if the epidemic is still ongoing. Each sample stores the sampled tree (`sampled.nwk`); `metadata.csv` also records the final sizes of the susceptible and recovered pools, which show how far the epidemic has depleted the susceptible pool.

```
output_dir: outputs/SIR-yaml
n_samples: {train: 100, val: 10, test: 10}
n_jobs: -1
seed: 42

context:
  R_0: {type: uniform, low: 1.5, high: 4}
  exit_rate: {type: uniform, low: 0.05, high: 0.2}
  s: {type: uniform, low: 0.2, high: 0.5}
  mu: exit_rate * (1 - s)
  psi: exit_rate * s
  init_S: {type: integers, low: 200, high: 1001}
  la: R_0 * exit_rate / init_S

model:
  spec: epidemic
  init_state: I
  init_pools: {S: init_S}
  events:
    - rule: {spec: infection, recipient_pool: S}
      hazard: {spec: constant, rate: la}
    - rule: {spec: recovery, destination_pool: R}
      hazard: {spec: constant, rate: mu}
    - rule: {spec: recovery, destination_pool: R, sample: true}
      hazard: {spec: constant, rate: psi}

stop_criterion: model.count_sampled_lineages() == 20

acceptance_criterion: model.count_current_lineages() > 0

logs:
  still_susceptible: model.pools["S"]
  recovered: model.pools["R"]

output_hooks:
  - model.history(model.get_sampled_lineages()).dump(output_dir / "sampled.nwk")
```

S3.2 SEIR: an incubation stage

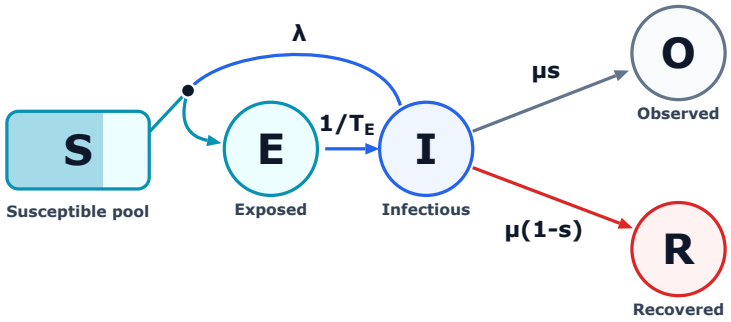


Figure S7: SEIR model.

SIR with infected hosts split into exposed (`E`: infected but not yet infectious) and infectious (`I`) lineages. Infections create `E` lineages, which become infectious after a mean time T_E; `I` lineages recover at rate $\mu = 1/T_I$, a fraction s of them sampled. As in SIR, $\lambda = R_0\mu/S_0$ makes R_0 the initial reproduction number. The trial stops at the twentieth sample, and is kept only if the epidemic is still ongoing. Each sample stores the sampled tree (`sampled.nwk`); `metadata.csv` also records the final sizes of the susceptible and recovered pools and the number of lineages still incubating.

```
output_dir: outputs/SEIR-yaml
n_samples: {train: 100, val: 10, test: 10}
n_jobs: -1
seed: 42

context:
  R_0: {type: uniform, low: 1.5, high: 4}
  T_E: {type: uniform, low: 1, high: 5}
  T_I: {type: uniform, low: 5, high: 20}
  mu: 1 / T_I
  s: {type: uniform, low: 0.2, high: 0.5}
  init_S: {type: integers, low: 200, high: 1001}
  la: R_0 * mu / init_S

model:
  spec: epidemic
  init_state: I
  init_pools: {S: init_S}
  events:
    - rule: {spec: infection, recipient_pool: S, source_state: I, child_states: [I, E]}
      hazard: {spec: constant, rate: la}
    - rule: {spec: recovery, source_state: I, destination_pool: R}
      hazard: {spec: constant, rate: mu * (1 - s)}
    - rule: {spec: recovery, source_state: I, destination_pool: R, sample: true}
      hazard: {spec: constant, rate: mu * s}
    - rule: {spec: migration, source_state: E, target_state: I}
      hazard: {spec: constant, rate: 1 / T_E}

stop_criterion: model.count_sampled_lineages() == 20

acceptance_criterion: model.count_current_lineages() > 0

logs:
  still_susceptible: model.pools["S"]
  recovered: model.pools["R"]
  still_incubating: model.count_current_lineages("E")

output_hooks:
  - model.history(model.get_sampled_lineages()).dump(output_dir / "sampled.nwk")
```

S3.3 SEIRS: waning immunity

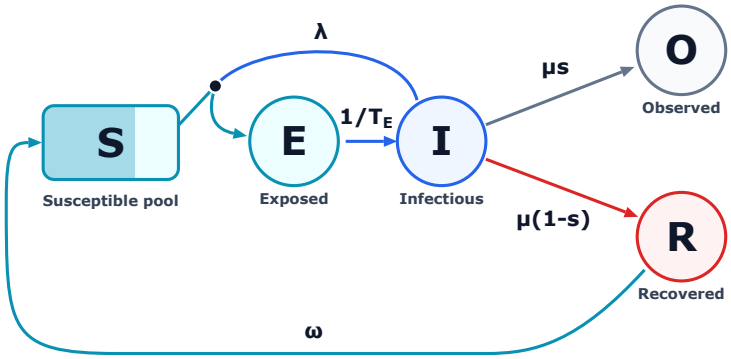


Figure S8: SEIRS model.

SEIR with a `flow` rule that returns each recovered host to the susceptible pool at rate $\omega = 1/T_R$, where T_R is the mean duration of immunity. Waning replenishes `S`, so the epidemic can persist instead of burning out. The trial stops at the hundredth sample, and is kept only if the epidemic is still ongoing. Each sample stores the sampled tree (`sampled.nwk`); `metadata.csv` also records the final sizes of the susceptible and recovered pools and the number of lineages still incubating.

```
output_dir: outputs/SEIRS-yaml
n_samples: {train: 100, val: 10, test: 10}
n_jobs: -1
seed: 42

context:
  R_0: {type: uniform, low: 1.5, high: 4}
  T_E: {type: uniform, low: 1, high: 5}
  T_I: {type: uniform, low: 5, high: 20}
  T_R: {type: uniform, low: 3 * T_I, high: 10 * T_I}
  omega: 1 / T_R
  mu: 1 / T_I
  s: {type: uniform, low: 0.2, high: 0.5}
  init_S: {type: integers, low: 100, high: 200}
  la: R_0 * mu / init_S

model:
  spec: epidemic
  init_state: I
  init_pools: {S: init_S}
  events:
    - rule: {spec: infection, recipient_pool: S, source_state: I, child_states: [I, E]}
      hazard: {spec: constant, rate: la}
    - rule: {spec: recovery, source_state: I, destination_pool: R}
      hazard: {spec: constant, rate: mu * (1 - s)}
    - rule: {spec: recovery, source_state: I, destination_pool: R, sample: true}
      hazard: {spec: constant, rate: mu * s}
    - rule: {spec: migration, source_state: E, target_state: I}
      hazard: {spec: constant, rate: 1 / T_E}
    - rule: {spec: flow, source_pool: R, destination_pool: S}
      hazard: {spec: constant, rate: omega}

stop_criterion: model.count_sampled_lineages() == 100

acceptance_criterion: model.count_current_lineages() > 0

logs:
  still_susceptible: model.pools["S"]
  recovered: model.pools["R"]
```

```
  still_incubating: model.count_current_lineages("E")

output_hooks:
  - model.history(model.get_sampled_lineages()).dump(output_dir / "sampled.nwk")
```

S4 Plugins

A plugin is an installable Python package that extends Phyloom with new functionality, such as a model or event rules, and registers it so that it can be used in YAML configurations. Once a plugin is installed, the `phyloom` command runs its configurations with all the usual machinery: parameter priors, stopping and acceptance criteria, parallel sample generation, logs, and output hooks. A plugin is needed when a process requires something the built-in models cannot express, such as state that belongs to individuals (identities, contact histories, or ages), waiting times that belong to a single lineage (non-exponential lifetimes), or events that trigger other events depending on the model's history (tracing the contacts of a sampled individual).

S4.1 Age-dependent branching

In a binary age-dependent branching process, also known as a Bellman-Harris process [29], every individual lives for a random lifetime drawn from a general distribution. At the end of its life, it either dies or splits into two new individuals, which start their own lifetimes from age zero. The classic application is cell proliferation, in which a cell that has just divided is unlikely to divide again immediately, and whose history can be recorded as a cell lineage tree by genetic lineage tracing [30]. This plugin simulates the multi-type version of the process:

1. Every lineage carries a discrete state, such as `A` or `B`.
2. When a lineage begins, it draws its lifetime from a distribution that depends on its state.
3. At the end of the lifetime, it dies with a state-dependent probability.
4. Otherwise, it is replaced by two descendants, whose states follow a configurable transition rule, and which start fresh lifetimes.

With non-exponential lifetimes, such as Gamma lifetimes with shape greater than one, the remaining lifetime of a lineage that has already lived for a while follows a different distribution than the lifetime of a newborn one. The chance that a lineage branches in the next instant therefore depends on its *age*.

Example

Two states, `A` and `B`, with Gamma lifetimes (fixed shapes above one, mean drawn per sample), state-specific death probabilities, and independent state switches of the two descendants at each split. Trials run up to a maximum time and are kept if at least a minimum number of lineages is still alive; each sample stores the full history tree and the number of extant lineages per state.

```
output_dir: outputs/ADB-multistate
n_samples: {train: 100, val: 10, test: 10}
n_jobs: -1
seed: 42

context:
  mean_lifetime_A: {type: uniform, low: 1.2, high: 1.8}
  mean_lifetime_B: {type: uniform, low: 0.7, high: 1.1}
  shape_A: 3.0
  shape_B: 4.0
  scale_A: mean_lifetime_A / shape_A
  scale_B: mean_lifetime_B / shape_B
```

```
  death_prob_A: {type: uniform, low: 0.35, high: 0.50}
  death_prob_B: {type: uniform, low: 0.45, high: 0.60}
  p_A_to_B: {type: uniform, low: 0.10, high: 0.35}
  p_B_to_A: {type: uniform, low: 0.10, high: 0.35}
  init_state: {type: choice, a: "['A', 'B']", p: "[0.5, 0.5]"}

model:
  spec: adb
  init_state: "{init_state}"
  lifetime_func: >-
    model.rng.gammavariate(
      shape_A if state == "A" else shape_B,
      scale_A if state == "A" else scale_B
    )
  death_prob: death_prob_A if state == "A" else death_prob_B
  get_child_states: >-
    model.rng.choices(
      ["A", "B"],
      weights=[1 - p_A_to_B, p_A_to_B] if state == "A" else
              [p_B_to_A, 1 - p_B_to_A],
      k=2
    )

max_time: 8.0

acceptance_criterion: model.count_current_lineages() >= 10

logs:
  extant_A: model.count_current_lineages("A")
  extant_B: model.count_current_lineages("B")
  extant_total: model.count_current_lineages()

output_hooks:
  - model.history().dump(output_dir / "history.nwk")
```

S4.2 Contact tracing

Infected individuals are lineages, each in a state, such as exposed or infectious. They transmit, creating new infected individuals, change state, and leave either unobserved or by being *sampled*, for example through a positive test. Which states exist, and which transitions are possible between them, is set by the events of the configuration. Contact tracing adds a response to each sample. The sampled individual becomes an *index case*, and its contacts are traced:

1. The contacts of an individual are the one that infected it and the individuals it infected, taken from most recent to least recent.
2. Only the first `max_notifications` contacts in that order are considered, or all of them if the limit is `null`.
3. Each considered contact that is still infected is independently notified with probability `notification_probability`.
4. A notified contact is sampled and removed at the same moment as the index case. Its own contacts are *not* traced further.

Samples therefore arrive in clusters of linked individuals with identical sampling times, and individuals in states that are never sampled directly, such as exposed individuals, can appear as tips once they are traced. Such data is useful for studying how tracing distorts phylodynamic inference [31], or for training methods that account for it [32].

Example

A BDEI epidemic with exposed (`E`) and infectious (`I`) individuals, parameterized by `R_0`, the mean infection time, the exposed fraction of it, and the sampling proportion. Each sampled individual notifies up to five contacts, each with probability `trace_probability`. Trials run until at least `n_leaves` individuals are sampled, and extinct epidemics are rejected; each sample stores the tree of the sampled individuals and the number of individuals still exposed and infectious.

```
output_dir: outputs/BDEI-CT
n_samples: {train: 100, val: 10, test: 10}
n_jobs: -1
seed: 42

context:
  R_0: {type: lognormal, mean: 1.0, sigma: 0.2}
  T: {type: lognormal, mean: 1.0, sigma: 0.3}
  f: {type: uniform, low: 0.05, high: 0.75}
  T_I: (1 - f) * T
  T_E: f * T
  epsilon: 1 / T_E
  mu: 1 / T_I
  p: {type: uniform, low: 0.05, high: 0.75}
  lambda_rate: R_0 * mu
  psi: mu * p
  death_rate: mu - psi
  init_state: {type: choice, a: "['E', 'I']", p: "[0.8, 0.2]"}
  n_leaves: {type: integers, low: 50, high: 250}
  trace_probability: {type: uniform, low: 0.5, high: 0.95}

model:
  spec: ct
  init_state: "{init_state}"
  notification_probability: trace_probability
  max_notifications: 5
  events:
    - rule: {spec: transmission, parent_state: I, child_state: E}
      hazard: {spec: constant, rate: lambda_rate}
    - rule: {spec: recovery, state: I, sample: false}
      hazard: {spec: constant, rate: death_rate}
    - rule: {spec: migration, source_state: E, target_state: I}
      hazard: {spec: constant, rate: epsilon}
    - rule: {spec: recovery, state: I, sample: true}
      hazard: {spec: constant, rate: psi}

stop_criterion: model.count_sampled_lineages() >= n_leaves

acceptance_criterion: model.count_current_lineages() > 0

logs:
  still_incubating: model.count_current_lineages("E")
  still_infectious: model.count_current_lineages("I")
  sampled_lineages: model.count_sampled_lineages()

output_hooks:
  - model.history(model.get_sampled_lineages()).dump(output_dir / "sampled.nwk")
```

S5 Computational performance

`Phyloom` advances a simulation one event at a time. At each step, every event type proposes a firing time from its current number of opportunities, and the event that fires draws the lineages it acts on uniformly at random from an indexed set, in constant time. For a fixed set of event types,

each step therefore costs constant time, independently of the population size, and a simulation of m events runs in $O(m)$ time. All lineages are kept in memory until the end of the simulation, so that event rules and output hooks can query them, and since each event creates or ends at most a few lineages, memory is also $O(m)$.

What m amounts to depends on the model. Forward-time models, such as birth–death and compartmental epidemics, simulate the full population, sampled and unsampled, so m is proportional to the number of simulated lineages. When a simulation stops once n tips are sampled and each lineage is sampled with probability p, this number scales as n/p, so both run time and memory grow linearly with n and with $1/p$. We evaluated this by generating trees with a growing number n of sampled tips under a constant-rate birth–death (BD) model (Section S1.1), in which infectious lineages branch at rate $\lambda = R_0\mu$ and are removed at rate $\mu = 0.4$, each removal being sampled with probability $p = 0.5$; $R_0 = \lambda/\mu = 3$ is the basic reproduction number, i.e. the expected number of lineages a lineage gives rise to before removal. We compared `phyloom` with `treesimulator` [31], a pure-Python simulator that also simulates the full history (Fig. S9). The run time and memory of `phyloom` grow close to linearly with n (fitted log–log slopes of 1.07 and 1.03), whereas the run time of `treesimulator` grows faster than linearly: `phyloom` is about 2.8-fold faster at 500 tips and 8.1-fold faster at 4 000 tips. In exchange, `phyloom` uses about 3-fold more memory, because each stored lineage is a full Python object that can be queried during and after the simulation. In practice, a tree costs roughly 0.08 ms and 8 kB per unit of n/p, so memory, not time, is the limiting factor: 16 GB of RAM allows n/p up to about two million. This bounds the population, rather than the tree, that forward-time models can handle. Backward-time coalescent models (Section S2) instead simulate only the ancestry of the samples, so m scales with the number of samples rather than with the population size, which makes them the better choice for sparsely sampled large populations.

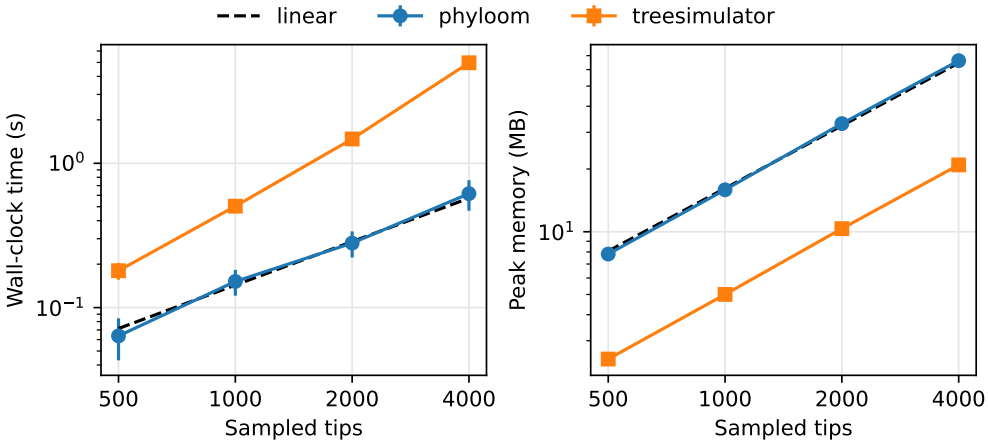


Figure S9: Wall-clock time (left) and peak memory (right) to generate a tree with a given number of sampled tips under a constant-rate BD model ($R_0 = 3$, $\mu = 0.4$, $p = 0.5$), for `phyloom` and `treesimulator` (mean ± s.d. over 5 replicates; log–log axes). The dashed line has slope 1, i.e. linear scaling, fitted to the `phyloom` values. Benchmarks were run single-threaded on a MacBook with an Apple M3 chip and 16 GB RAM, with Python 3.10.

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